ITA03-MOBILE COMPUTING ASSIGNMENT-2

ASSIGNMENT 2 – CO2

CO2 Coverage: Mobile Telecommunication Systems; Multiple Access Techniques (SDMA, TDMA, FDMA, CDMA, OFDMA); GSM Architecture, Services, Radio Interfaces, Localization, Calling, Handover & Security; Advanced GSM Extensions (GPRS, HSCSD, DECT).

Nature of Assignment: Individual
Duration: 1 Week
Total Marks: 50
Submission Date: 4/11/2026

Q1 – Mobile Communication or Mobile Computing? 5 Marks Problem Context –

Field-Service Technology Audit

A utility company equips its field staff with: (a) a ruggedised tablet fetching work orders over cellular data, (b) a laptop plugged into Ethernet at different offices, (c) a fixed wireless router at a remote substation, (d) a push-to-talk handset, and (e) a tablet app that records meter readings offline and synchronises later.

Your Challenge

Investigate whether "wireless" and "mobile" mean the same thing, and whether mobile communication and mobile computing are the same thing.

Before analysis:

- Define at least four of your own classification criteria (e.g., user mobility, wireless link, tolerance of disconnection, location/context awareness, device resource limits).
- Predict how each of the five cases is classified (fixed-wired, nomadic, fixed wireless, mobile) and whether it is mainly a mobile-communication or a mobile computing problem, giving reasons.
- Design one additional ambiguous case of your own to test your criteria. **Test**

Your Reasoning

Build a small simulation of case (e) with link availability of 90%, 60% and 30%. Compare

two strategies: send immediately (fail if disconnected) vs store-and-forward queue. Record updates delivered, delayed and lost, and mean delivery delay.

Inquiry

Which characteristics of mobile computing are exposed by your results that a wireless link alone does not solve? Which device categories need application-level changes?

Evidence Expected: Own classification criteria, reasoned predictions, an engineered ambiguous case, simulation counts for both strategies, and a justified conclusion.

Q2 – Sharing the Channel – FDMA vs TDMA Under Different Traffic 6 Marks

Problem Context – Hostel-Campus Voice and Telemetry

A cell may use a 2 MHz band per direction. Option 1 is FDMA with 25 kHz channels. Option 2 is TDMA with 200 kHz carriers, each carrying 8 time slots. Both offer 80 nominal channels.

Your Challenge

Investigate whether equal nominal capacity means equal usable capacity.

- Estimate usable channels after guard bands, guard times, synchronisation and control overheads (state and justify your values).
- Predict which scheme gives better utilisation for (i) continuous voice, (ii) bursty data with about 10% duty cycle, and (iii) a 70/30 voice–data mix, and analyse why.

Design three contrasting traffic datasets, including one edge case of your own (e.g., very short calls or synchronised arrivals). Simulate dedicated-channel allocation under both schemes and record utilisation, blocking and wasted capacity.

Inquiry

Where do the two schemes diverge, and what causes it?

Design Decision

At a campus telemetry site, 90% of devices send a 200-byte report per minute. Is dedicated FDMA/TDMA channel allocation justified? Defend your answer.

Evidence Expected: Capacity calculations with overheads, reasoned predictions, three designed traffic conditions, utilisation/blocking evidence and a justified decision.

Q3 – CDMA – Orthogonality, Spreading and the Near-Far Problem 7 Marks

Problem Context – Crowded Stadium Uplink

Several handsets transmit simultaneously in the same band to one base station, each spreading its data with its own code.

Your Challenge

Design a direct-sequence CDMA uplink simulation. Determine how data bits are spread, how the receiver despreads and decides, and what the processing gain is.

Before executing, predict the bit-error behaviour as the number of users grows for: (i) orthogonal (e.g., Walsh) codes with synchronous users, (ii) random non-orthogonal codes of the same length, and (iii) orthogonal codes where one user transmits at ten times the power. analyse your reasoning.

Implement and test with at least three code lengths (e.g., 8, 16, 64). Record bit errors over a stated number of bits, correlation values and processing gain.

Inquiry

- Why do orthogonal codes still fail when users are unsynchronised or powers are unequal?
- Why is power control essential in CDMA?
- Estimate the number of simultaneous users supported for a bit-error target that you define.

Evidence Expected: A working DS-CDMA simulation, predictions made before execution, bit-error evidence for three code lengths and three conditions, and a reasoned explanation of the results.

Q4 – OFDMA – Giving Each User the Subcarriers They Can Use Best 6 Marks

Problem Context – Lecture-Hall Downlink

Eight users share 64 subcarriers. Each user's channel quality varies across subcarriers (frequency-selective fading).

Your Challenge

- Before designing, determine how subcarrier spacing relates to symbol duration, why a cyclic prefix is needed, and verify orthogonality of at least three subcarriers numerically (FFT or inner product).
- Design three allocation approaches: fixed contiguous blocks, interleaved round-robin, and channel-aware best-subcarrier.
    - Predict total throughput and fairness (e.g., Jain's index) of each, with reasons.

Implement using generated fading gains with fixed seeds over at least 1,000 channel realisations. Record sum throughput, per-user throughput and fairness.

Inquiry

Why can maximising total throughput starve some users? How does the number of users change the benefit of channel-aware allocation?

Design Decision

Select an allocation for a lecture hall in which every student must receive at least a minimum rate that you define and justify.

Evidence Expected: Numerical orthogonality verification, predictions, three allocation schemes implemented, throughput and fairness results, and a justified allocation choice.

Q5 – SDMA – Reusing the Same Channel in Different Directions 6 Marks

Problem Context – Highway Cell With Directional Antennas

Vehicles along a highway use the same frequency and time slot. The base station can serve them with an omni antenna, a 120° sector antenna, or an N-element uniform linear array steered toward each user.

Your Challenge

- Derive the array factor of an N-element uniform linear array and predict how main-lobe width and side-lobe level change with N.
- Define an SIR threshold and predict the minimum angular separation at which two users can share the same resource.
- Compute patterns for N = 2, 4, 8, 16 and evaluate the SIR at user B when the beam is steered at user A, for separations of 5°–60°. Record beamwidth, side lobe level and the minimum separation meeting your threshold.

Inquiry

What limits SDMA in practice (angular spread, clustered users)? Why can two users in nearly the same direction not be separated?

Design Decision

At a toll plaza most vehicles arrive in the same direction. Would you rely on SDMA there? Defend your decision.

Evidence Expected: Array-pattern derivation, predictions, computed patterns and SIR results across angular separations, and a reasoned explanation of SDMA limits.

Q6 – Choosing Multiple-Access Techniques for Real Deployments 7 Marks Problem

Context – Four Deployments

Choose access methods for: (1) wide-area cellular voice, (2) a jam-resistant tactical link, (3) a high-rate stadium downlink, (4) a network of 5,000 low-rate sensors sending small reports.

Your Challenge

- Define at least five weighted criteria and justify the weights yourself. ● Rate each of SDMA, TDMA, FDMA, CDMA and OFDMA for each deployment using your own Q2–Q5 evidence wherever possible.
- Choose a primary technique and, where useful, a hybrid for each deployment. ● Perform a sensitivity check by changing weights by ±20%: does any choice change?

Inquiry

Why do real systems combine techniques (for example FDMA with TDMA)? Which of your choices is least robust, and why?

Evidence Expected: Independently justified criteria and weights, ratings traceable to your own Q2–Q5 evidence, sensitivity analysis and defended choices.

DAY-4 DESIGN CHALLENGE

Q7 – Diagnose, Redesign and Validate 13 Marks Review the evidence you generated

while working on Q1–Q6.

Select **ONE** solution that demonstrates a significant limitation as its problem size or operating requirement increases. Examples of limitations you may find in this variant:

- Dedicated FDMA/TDMA channels sit idle under bursty traffic.
- A CDMA link collapses under power imbalance or loss of synchronisation. ● Best-subcarrier OFDMA allocation starves weak users.
- SDMA fails when users are angularly clustered.

You are not permitted simply to replace the solution with a more advanced technology or standard and state that it is better. Your task is to understand the problem first and then develop an appropriate redesign.

A. Diagnose the Bottleneck 2 Marks

- Where most capacity, signalling or computational effort is being spent. ● Which counted quantity grows significantly as load or size increases. ● Whether resources are idle, wasted or repeated unnecessarily. ● Which characteristic of the original design causes the limitation.

Evidence Expected: A precise diagnosis supported by evidence from your original implementation and analysis.

B. Define the Design Objective 2 Marks

- State what your redesign must achieve before proposing it (e.g., reduce wasted capacity, reduce signalling messages, avoid unnecessary resource holding, reuse information already available).
- Specify a measurable objective wherever possible.

Evidence Expected: A clearly stated, measurable objective derived directly from the diagnosed limitation.

C. Design the Revised Strategy 4 Marks

- What you have changed and what property or information the new design exploits.
- Which operations or resource usages are avoided or reduced. ● Whether additional state, preprocessing or complexity is introduced, and any new trade-offs.
- Provide pseudocode, a flowchart or a message-sequence design. ● Marks are based on the quality of reasoning, not merely on naming a known technique.

Evidence Expected: A coherent redesign directly addressing the bottleneck, with a design representation and technical justification.

D. Predict Before Implementing 1 Mark

- Predict how the redesigned solution should differ from the original and in which measure you expect improvement.

Evidence Expected: A clear, testable prediction made before experimental validation. **E.**

Implement and Validate the Redesign 2 Marks

- Implement a prototype or simulation and compare Original → Redesigned under the same or equivalent test conditions.
- Use counted measures (e.g., slots used, messages, blocking, bit errors, throughput, fairness); run-time is supporting evidence only.

Evidence Expected: Comparable experimental evidence for the original and redesigned solutions.

F. Defend the Final Design 2 Marks

- Was the prediction supported, and what was the magnitude of improvement? ● What new costs or trade-offs appeared, and when might the original still be preferable?
- A redesign that does not outperform the original is not automatically unsuccessful if you correctly analyse why the hypothesis failed.

Evidence Expected: A critical, evidence-based evaluation and a defensible final engineering conclusion.

ASSIGNMENT-2 – ANSWERS

Planning assumptions used throughout the answers: The numerical values in the calculations are based on the assignment's stated data. Where the question does not specify a parameter, an explicit illustrative assumption is stated. Simulation result examples are labelled as illustrative and must not be presented as measured results unless the corresponding code is run.

ANSWER – Q1: Mobile Communication or Mobile Computing?

1. Classification criteria used

User mobility: whether the user/device changes physical location while working.

Connection type: wired, fixed wireless, or cellular/Wi-Fi wireless.

Disconnection tolerance: whether the task can continue without a network and synchronise later.

Context awareness: whether location, time, or changing network conditions affect the application.

Main purpose: communication transports voice/data; computing processes, stores, or acts on data.

2. Classification of the five cases

Case	**Mobility / link classification**	**Main problem**	**Reason**
(a) Rugged tablet using cellular work orders	Mobile; wireless	Both, mainly mobile computing	The tablet moves between field locations; the application must handle changing coverage and work-order data.
(b) Laptop plugged into Ethernet at different offices	Nomadic; fixed-wired while connected	Mainly mobile computing	The user moves between offices, but each session uses a wired link. Mobility is not the same as wireless access.
(c) Fixed wireless router at a remote substation	Fixed wireless	Mainly mobile communication/networking	The equipment is stationary; the radio link replaces a cable, but the device is not mobile.

(d) Push-to-talk handset	Mobile; wireless	Mainly mobile communication	The key service is real-time voice/group communication while the user moves.
(e) Tablet stores meter readings offline and synchronises later	Mobile; intermittently connected	Mainly mobile computing	Local storage, queueing, conflict handling, and delayed synchronisation are application responsibilities.

3. Additional ambiguous case

A technician carries a tablet around a factory, but it connects only to a wall-mounted Wi-Fi access point and saves inspection results locally when Wi-Fi is unavailable. It is mobile because the user moves, wireless when connected, and also a mobile-computing case because the application must tolerate outages.

4. Simulation assumptions and step-by-step calculation

Assume 100 meter updates are generated uniformly during a test. Link availability is the probability that an update finds the connection available. For the simple immediate-send strategy, a failed attempt is counted as lost. For store-and-forward, disconnected updates remain queued and are delivered at the next available opportunity. To make delay reproducible, assume a 1-minute check interval and that an update generated during an outage waits, on average, half of the unavailable portion of the interval. The table below is an illustrative model calculation, not a measurement from a physical network.

Immediate-send delivered = 100 × availability. Lost = 100 − delivered. Delayed = 0 because failed updates are discarded.

Store-and-forward eventually delivers all 100 updates if the queue has enough storage and connectivity returns; delayed updates = 100 × (1 − availability).

Illustrative mean delay for queued updates = (1 − availability) × 0.5 minute. This simple delay assumption can be replaced by measured delay in a real simulation.

Link availability	**Immediate delivered**	**Immediate lost**	**Queue delivered**	**Queue delayed**	**Illustrative mean queue delay / update**
90%	90	10	100	10	0.05 min (3 s)

60%	60	40	100	40	0.20 min (12 s)
30%	30	70	100	70	0.35 min (21 s)

5. Findings and conclusion

A wireless link alone does not guarantee reliable application completion. Mobile computing also needs local storage, retry logic, timestamps, duplicate prevention, conflict resolution, battery-aware operation, and clear user feedback. Cases (a) and (e) need application-level changes; (b) may need session continuity and offline support when moving between offices. Case (c) primarily needs reliable fixed-wireless networking, while (d) needs handover, low latency, and voice-session continuity. Therefore, wireless and mobile are not synonyms, and mobile communication is a component of mobile computing rather than the same concept.

ANSWER – Q2: FDMA vs TDMA Under Different Traffic

1. Nominal capacity calculation

Total bandwidth per direction = 2 MHz = 2,000 kHz.

FDMA nominal channels = 2,000 kHz ÷ 25 kHz/channel = 80 channels.

TDMA nominal carriers = 2,000 kHz ÷ 200 kHz/carrier = 10 carriers.

TDMA nominal time slots = 10 carriers × 8 slots/carrier = 80 slots.

2. Assumed overhead and usable capacity

The question does not provide exact guard-band, guard-time, synchronisation, or control values. For a transparent comparison, assume FDMA loses 8% of nominal capacity to guard bands/control, while TDMA loses 10% to guard time, synchronisation, and control. These are assignment assumptions, not standard-defined values.

FDMA usable channels = $80 \times (1 - 0.08) = 73.6 \approx 74$ channels.

TDMA usable slots = $80 \times (1 - 0.10) = 72$ slots.

Although both begin at 80 nominal channels, usable capacities differ because their overheads differ.

Scheme	**Nominal capacity**	**Assumed overhead**	**Estimated usable capacity**
FDMA	80 channels	8%	$80 \times 0.92 = 73.6 \approx 74$
TDMA	80 slots	10%	$80 \times 0.90 = 72$

3. Traffic predictions

Traffic condition	Expected better utilisation	Reason
Continuous voice	FDMA or TDMA, depending on implementation	Voice needs a steady rate; dedicated FDMA channels avoid slot scheduling, while TDMA can share carrier hardware efficiently.
Bursty data, 10% duty cycle	TDMA if slots can be reassigned; otherwise neither is efficient	A dedicated channel sits idle about 90% of the time when the source is inactive.
70% voice / 30% data	TDMA with scheduling, or a hybrid	Voice can receive regular slots while data uses available slots; strict fixed allocation wastes capacity when traffic varies.

4. Three designed traffic datasets and illustrative allocation results

For a small reproducible comparison, assume 72 usable resource units for TDMA and 74 for FDMA. Each active call/report occupies one resource unit. Utilisation = occupied resource units ÷ usable units × 100%. Blocking = requests rejected because all units are occupied. The figures below are illustrative scenario counts chosen to demonstrate the method; replace them with counts from an actual simulation before claiming measured results.

Dataset	Offered load pattern	FDMA utilisation / blocked	TDMA utilisation / blocked	Interpretation
A: steady voice	60 continuously active calls	60/74 = 81.1%; 0 blocked	60/72 = 83.3%; 0 blocked	Both work well for steady demand.
B: bursty telemetry	8 active reports out of 80 registered devices at a time	8/74 = 10.8%; 0 blocked	8/72 = 11.1%; 0 blocked	Most dedicated capacity remains idle.
C: short synchronised burst	80 requests arrive together	74/74 = 100%; 6 blocked	72/72 = 100%; 8 blocked	A sudden peak exceeds available resources;

				capacity and admission policy matter.

5. Campus telemetry decision

Report size = 200 bytes × 8 = 1,600 bits per report.

At one report per minute, average payload rate per device = 1,600 ÷ 60 = 26.67 bit/s, before headers and retransmissions.

For 5,000 devices, aggregate payload rate = 5,000 × 26.67 = 133,350 bit/s ≈ 133.35 kbit/s.

Only 10% of devices transmit in a given instant if activity is asynchronous; this is still a low-rate, bursty traffic pattern compared with continuously reserved voice channels.

Conclusion: Dedicated FDMA/TDMA allocation is generally not justified for thousands of low-rate telemetry devices because resources would remain idle most of the minute. A scheduled or contention-based shared channel, with batching and retry control, is more suitable. Dedicated resources remain useful for continuous voice or latency-sensitive traffic.

ANSWER – Q3: CDMA, Spreading and the Near–Far Problem

1. DS-CDMA operation

Map each data bit to bipolar form: bit 1 → +1 and bit 0 → −1.

Assign each user a spreading code of length L, containing +1 and −1 chips.

Spread one data bit by multiplying it by all L chips of that user's code. One bit therefore becomes L chips.

Add the chip sequences from simultaneous users to form the received signal.

At the receiver, multiply the received chips by the desired user's code and average/sum across L chips. A positive result decides bit 1; a negative result decides bit 0.

2. Processing gain calculation

Processing gain in linear form is Gp = chip rate ÷ bit rate = code length L. In decibels, Gp(dB) = 10 log10(L).

Code length L	**Linear processing gain**	**Step-by-step dB calculation**	**Approximate gain**
8	8	10 log10(8)	9.03 dB
16	16	10 log10(16)	12.04 dB
64	64	10 log10(64)	18.06 dB

3. Correlation and predictions before testing

For two code vectors a and b, normalised correlation is $\rho = (a \cdot b)/L$. Orthogonal codes have $a \cdot b = 0$, so $\rho = 0$ when aligned and synchronised. Random codes usually have non-zero correlation, causing multi-user interference.

Condition	**Prediction as user count grows**	**Reason**
Synchronous orthogonal codes, equal power	Lowest bit-error rate while code orthogonality is preserved	Cross-correlation ideally equals zero.
Random non-orthogonal codes	Bit errors increase sooner	Non-zero cross-correlation adds interference to the desired user's despread signal.
Orthogonal codes, one user at 10× power	Near–far interference may cause errors for weaker users	Timing errors, imperfect channel estimation, and non-ideal orthogonality let strong-user energy leak into other users.

4. Small reproducible simulation plan

Use a fixed random seed (for example 42), 10,000 information bits per user, and code lengths L = 8, 16, 64. Run each of the three conditions separately. Record total transmitted bits, error bits, BER = errors ÷ tested bits, code cross-correlations, and processing gain. For a fair comparison, keep the same bit sequence and noise setting between the original and modified cases.

5. Worked correlation example

Let a = [+1, +1, −1, −1] and b = [+1, −1, +1, −1]. Their dot product is (1×1) + (1×−1) + (−1×1) + (−1×−1) = 1 − 1 − 1 + 1 = 0. Therefore $\rho = 0/4 = 0$: these two codes are orthogonal when aligned.

6. Capacity estimate and conclusion

Define the target as BER ≤ 1%. The maximum simultaneous users cannot be calculated from code length alone; it depends on received SNR, synchronisation, code cross-correlation, channel conditions, and power control. A defensible experiment should increase users one at a time and report the largest count for which measured BER remains ≤ 1%. Power control is essential because the base station should receive users at approximately comparable power; otherwise, a nearby strong user can overwhelm a distant weak user.

ANSWER – Q4: OFDMA Subcarrier Allocation

1. Subcarrier spacing and symbol duration

For an OFDM system with N equally spaced orthogonal subcarriers, $\Delta f = 1/Tu$, where Δf is subcarrier spacing and Tu is the useful symbol duration. For example, if $\Delta f = 15$ kHz, $Tu = 1/15{,}000 = 66.67$ μs.

A cyclic prefix (CP) copies the end of the OFDM symbol to its beginning. It reduces inter-symbol interference caused by multipath delay, provided the CP is longer than the relevant channel delay spread. It adds overhead, so the total symbol duration is $Ts = Tu + Tcp$.

2. Numerical orthogonality verification

For subcarriers k and m over the useful symbol interval, the inner product is proportional to $\int_0^{Tu} \exp(j2\pi(k-m)t/Tu)dt$. If $k \neq m$ and the interval is exactly Tu, the positive and negative rotations cancel and the integral is zero. For $k = m$, the integral equals Tu. For example, with $Tu = 1$ ms and $\Delta f = 1$ kHz, subcarriers at 0, 1 kHz and 2 kHz complete integer cycles during the interval; their pairwise normalised inner products are ideally 0, 0 and 0.

Pair	**Frequency difference**	**Cycles across Tu**	**Ideal normalised inner product**
k=0, m=1	1 kHz	1	0
k=0, m=2	2 kHz	2	0
k=1, m=2	1 kHz	1	0

3. Allocation approaches

Approach	**Rule**	**Expected benefit**	**Risk**
Fixed contiguous blocks	Give each user 8 adjacent subcarriers (64 ÷ 8)	Simple and predictable; each user gets equal resource count	A user may receive a block with poor fading gains.
Interleaved round-robin	Allocate every eighth subcarrier to each user	Spreads each user's resources across frequency	Does not necessarily assign the best channel to the user.
Channel-aware best-subcarrier	Assign each subcarrier to the user with the highest channel quality	Can maximise sum throughput	Weak users may receive few resources and fairness can fall.

4. Step-by-step throughput and fairness method

Generate a 8 × 64 channel-gain matrix for each channel realisation using a fixed seed.

For each allocation, compute each assigned subcarrier rate using $r = \log_2(1 + SNR \times gain)$ bits/s/Hz (or the same chosen modulation table for all schemes).

Sum assigned rates to obtain total throughput and sum each user's assigned rates to obtain per-user throughput.

Compute Jain's fairness index: $J = (\Sigma x_i)^2 / (n \times \Sigma x_i^2)$, where x_i is user i's throughput and $n = 8$. $J = 1$ means perfectly equal rates.

5. Illustrative result table

The following values are a worked example to show how to report the results; actual values must be produced by running the fixed-seed simulation over at least 1,000 realisations.

Allocation	**Example sum throughput (relative units)**	**Example Jain fairness**	**Interpretation**
Fixed contiguous	42	0.96	Fair and predictable, but may miss good subcarriers.
Interleaved round-robin	46	0.98	Frequency diversity improves robustness and equal resource counts help fairness.
Channel-aware best-subcarrier	58	0.62	Highest sum throughput, but strong-channel users can dominate.

6. Design decision

For a lecture hall, use proportional-fair or minimum-rate-constrained channel-aware allocation. First reserve enough subcarriers to meet each student's minimum rate, then distribute remaining subcarriers according to channel quality. This balances throughput and fairness. Pure maximum-throughput allocation is unsuitable if it repeatedly starves weak users.

ANSWER – Q5: SDMA and Directional Antennas

1. Uniform linear array (ULA) array factor

For N equally spaced antenna elements with spacing d, wavelength λ, and steering angle θ_0, a normalised array factor can be written as $AF(\theta) = |\sin(N\psi/2) / (N \sin(\psi/2))|$, where $\psi = (2\pi d/\lambda)(\sin\theta - \sin\theta_0)$. At $\psi = 0$, the limiting value is 1 (the main-beam maximum).

2. Prediction before calculation

As N increases, the main lobe becomes narrower and angular discrimination improves.

The ideal uniform-amplitude array's first sidelobe is approximately −13 dB relative to the main lobe; increasing N narrows the beam but does not automatically eliminate sidelobes.

Closely spaced users have similar steering angles, so the beam may not separate them well.

An omni antenna provides little angular discrimination; a 120° sector antenna provides coarse separation; a steered array can provide finer separation.

3. Worked numerical setup

Assume half-wavelength spacing $d = \lambda/2$ and steer the array to user A at $\theta_0 = 0°$. Then $\psi = \pi \sin\theta$. For a user B at angular separation θ, the normalised response is $AF(\theta) = |\sin(N\pi \sin\theta/2) / (N \sin(\pi \sin\theta/2))|$. Approximate power response is $|AF|^2$.

To evaluate interference, define SIR threshold = 10 dB. If user A's transmitted power is normalised to 1 and the response toward B is $|AF(\theta)|^2$, a simple single-interferer illustration is SIR_B ≈ desired power / leakage power. A full multiuser SIR also requires the desired user's own beam gain, other-user powers, and noise.

Array elements	**Approx. broadside HPBW**	**Ideal first sidelobe**	**Expected angular separation**
N = 2	≈ 60°	≈ −13 dB	Largest; broad beam has weak separation
N = 4	≈ 29°	≈ −13 dB	Better than 2 elements
N = 8	≈ 14°	≈ −13 dB	Can separate more closely spaced users
N = 16	≈ 7°	≈ −13 dB	Narrowest main beam among the cases

The beamwidth figures are approximate broadside estimates, not exact simulation output. Exact HPBW, sidelobe level, and minimum separation must be computed from sampled patterns for 5°, 10°, …, 60° and checked against the 10 dB threshold. Record the first separation where the measured SIR reaches or exceeds 10 dB.

4. Why SDMA has limits

SDMA is limited by angular spread, multipath, array calibration, finite antenna count, mutual coupling, and users clustered in the same direction. Two users nearly aligned in angle produce similar spatial signatures; the receiver cannot reliably distinguish them using direction alone.

5. Toll-plaza decision

At a toll plaza, most vehicles approach from nearly the same direction, so relying only on SDMA is risky. Use sector coverage and time/frequency scheduling, with SDMA as an additional gain when vehicles are sufficiently separated in angle. The hybrid is more robust than a beam-only solution.

ANSWER – Q6: Selecting Multiple-Access Techniques

1. Weighted decision criteria

Rate each method from 1 (poor) to 5 (excellent). Weights total 100%. The weights are design assumptions for this assignment and should be adjusted if the deployment priorities change.

Criterion	**Weight**	**Reason**
Spectral efficiency	25%	Important when bandwidth is limited.
Latency / real-time support	20%	Important for voice and interactive services.
Robustness to interference	20%	Important in tactical and crowded radio environments.
Scalability / many devices	20%	Important for stadiums and 5,000 sensors.
Fairness / predictable service	15%	Avoids starving users and supports service guarantees.

2. Rating matrix (1 = poor, 5 = excellent)

Technique	**Spectral efficiency**	**Latency**	**Interference robustness**	**Scalability**	**Fairness**
SDMA	4	4	3	4	3
TDMA	3	4	3	3	4
FDMA	2	4	3	2	4
CDMA	4	3	4	4	3
OFDMA	5	4	3	5	4

3. Weighted-score calculation example

For OFDMA: score = 0.25(5) + 0.20(4) + 0.20(3) + 0.20(5) + 0.15(4) = 1.25 + 0.80 + 0.60 + 1.00 + 0.60 = 4.25 out of 5.

Technique	**Weighted score / 5**
SDMA	0.25(4)+0.20(4)+0.20(3)+0.20(4)+0.15(3) = 3.65
TDMA	0.25(3)+0.20(4)+0.20(3)+0.20(3)+0.15(4) = 3.35
FDMA	0.25(2)+0.20(4)+0.20(3)+0.20(2)+0.15(4) = 2.90
CDMA	0.25(4)+0.20(3)+0.20(4)+0.20(4)+0.15(3) = 3.65
OFDMA	4.25

4. Choices for four deployments

Deployment	**Primary choice**	**Useful hybrid**	**Reason**
Wide-area cellular voice	TDMA/FDMA-style resource structure or OFDMA in modern systems	FDMA + TDMA / OFDMA + SDMA	Regular resource allocation supports voice quality and frequency reuse.
Jam-resistant tactical link	CDMA / spread-spectrum approach	Frequency hopping and coding where available	Spreading can make interference harder to exploit, although it does not make a link immune to jamming.
High-rate stadium downlink	OFDMA	OFDMA + SDMA/MIMO	Subcarriers can be assigned based on channel quality and spatial streams can reuse resources.
5,000 low-rate sensors	Scheduled/shared access rather than	TDMA scheduling with aggregation and sleep cycles	Small, infrequent reports make permanent

	one dedicated channel per device		dedicated allocation inefficient.

5. Sensitivity check

Change each criterion weight by ±20%, then renormalise weights to 100% and recalculate scores. OFDMA is expected to remain a strong choice for high-rate stadium traffic because it scores highly in spectral efficiency and scalability. The tactical-link decision is less robust: increasing the robustness weight favours CDMA/spread spectrum, while prioritising deterministic latency may favour scheduled TDMA. The sensor decision is also sensitive to device synchronisation, battery life, and traffic burstiness.

Conclusion: Real systems combine methods because no single technique optimises every requirement. FDMA separates frequency bands, TDMA separates time, CDMA separates codes, OFDMA allocates orthogonal subcarriers, and SDMA reuses resources spatially. Hybrid designs balance capacity, latency, interference, and fairness.

ANSWER – Q7: Day-4 Design Challenge – Diagnose, Redesign and Validate

Selected limitation: Dedicated FDMA/TDMA channels waste capacity when telemetry traffic is bursty.

A. Diagnose the bottleneck (2 marks)

A dedicated channel is reserved even when a sensor has no report to send.

For a 10% duty-cycle source, the channel is idle for approximately 90% of the time.

As the number of sensors increases, the number of reserved channels grows even though only a small subset is transmitting at any moment.

The limitation is fixed resource reservation rather than the radio bandwidth alone.

B. Define the design objective (2 marks)

Objective: reduce idle reserved resource units by at least 50% during bursty traffic while delivering at least 99% of generated reports within the selected reporting deadline. The deadline and delivery target are design assumptions and should be checked against the application requirements.

C. Revised strategy (4 marks)

Use a shared scheduled pool. Sensors keep reports locally, request or receive a short transmission opportunity, send the report, and release the resource immediately. Batch multiple readings when permitted. Add sequence numbers and acknowledgements so retries do not create duplicate records.

Pseudocode

```
for each reporting interval:
    collect reports from sensors
    place each report in a local queue with a sequence number
    announce or assign available shared transmission slots
    while slots remain and queued reports exist:
```

```
        select a queued report
        transmit report in an assigned slot
        if acknowledgement received:
            remove report from queue
        else:
            retain report for retry
    record delivered, delayed, lost, occupied_slots, and idle_slots
```

Operations reduced: permanent reservation for silent devices and idle channel holding. Additional costs: scheduling messages, queue memory, acknowledgement overhead, and possible delay while waiting for a slot. Dedicated channels may still be preferable for continuous voice or strict low-latency services.

D. Prediction before implementation (1 mark)

Prediction: under 10% duty-cycle traffic, shared scheduling should reduce reserved-but-idle resource units by more than 50%, while maintaining delivery through queueing. It may add a small scheduling delay compared with an immediately available dedicated channel.

E. Illustrative validation calculation (2 marks)

Assume 100 resource units are available per reporting interval and 10 sensors are active on average. In a dedicated design, if 100 sensors each reserve one unit, 100 units are reserved and 90 are idle. In a shared design, 10 units are occupied by the active reports; if 2 extra units are used for scheduling/control, total used is 12.

Measure	**Original dedicated allocation**	**Redesigned shared pool**
Reserved/used units	100 reserved	10 data + 2 control = 12 used
Idle data units	90	0 of the 10 data slots assigned; remaining capacity stays available to other traffic
Data utilisation against 100 units	10/100 = 10%	10/100 = 10% payload; only 12% including control
Resource reservation reduction	Baseline	(100 − 12) ÷ 100 × 100 = 88% fewer reserved/used units
Report delivery	Depends on assigned channel availability	Queued reports delivered when slots and acknowledgements are available

The 88% figure is an illustrative arithmetic comparison under the stated assumptions, not a measured test result. A proper validation run should use identical generated arrivals and fixed random seeds for both designs and export CSV columns for arrival time, delivery time, retries, slot count, blocked reports, and lost reports.

F. Defend the final design (2 marks)

The redesign addresses the root cause by sharing resources among intermittent senders rather than reserving a channel for each device. The predicted reduction in reservation is supported by the example calculation. The trade-off is extra scheduling and queueing delay; therefore, shared access is suitable for periodic telemetry, while dedicated or guaranteed slots remain useful for continuous voice and emergency traffic.

Overall conclusion

The assignment shows that multiple-access design is a trade-off among capacity, reliability, fairness, delay, and complexity. Capacity calculations and simulations should state their assumptions, use fixed seeds, and report counted measures. Illustrative tables in this document demonstrate the calculation method; they should not be represented as actual measured results unless the corresponding simulations are run and the outputs are recorded.

Mark Distribution

Question	Assessed Area	Marks
Q1	Mobile communication vs. mobile computing	5
Q2	FDMA and TDMA	6
Q3	CDMA	7
Q4	OFDMA	6
Q5	SDMA	6
Q6	Comparison of SDMA, TDMA, FDMA, CDMA, OFDMA	7
Q7	Day-4 Design Challenge – Diagnose, Redesign & Validate	13
TOTAL		50

Submission Requirements (Digital)

- Assignment report (PDF) containing the required evidence for Q1–Q7. No handwritten work.
- Source code for all implementations/simulations (GitHub repository link with README and fixed random seeds).
- Raw experimental data (CSV) used to produce your results.
- Tables and graphs used in your analysis.
- Counted quantities (slots, messages, bits, blocked calls, codes, throughput) are primary experimental evidence. Execution time may be reported as supporting evidence but must not be the sole basis for conclusions.
- Standards parameters (frame durations, rates, header sizes, etc.) must be obtained from cited references; state assumptions wherever a value is assumed.

Assessment Verification

You may be asked to analyse any part of your design, code, test-data design, experimental results, predictions, analysis and Design Challenge in a brief verification interpretion/viva. Disclose any AI or reference tools used. You must be able to analyse and defend the evidence as your own work.

Overall Assessment Expectation

This assignment does not primarily assess your ability to reproduce textbook descriptions of mobile systems. It assesses your ability to:

Predict → Design → Implement → Measure → Analyse → Evaluate → Diagnose → Redesign → Validate

Your conclusions and design decisions must be supported by your own reasoning and experimental evidence.